ENME 561/661

Lab 3



Peer Review & Grading Procedure

- **Submission of Work**
  - Each group hands in one printed copy of their experimental procedure to the TA at the start of class.
  - Ensure your **first names** and **Group Number (from D2L)** are written at the top of the page
- **Distribution**
  - The TA will assign each printed file to a different group for review.
  - Groups are **not allowed to discuss verbally** or request verbal clarification from the authors of the procedure.
- **Peer Review Task**
  - The reviewing group carefully reads the assigned procedure.
  - They complete the peer review using the **Microsoft Form,** which includes:
    - Rubric scoring (0–2 points per criterion)
    - One short written comment (strengths + suggestions for improvement)
- **Submission of Reviews**
  - Each group submits their review **only through the MS Form**.
- **Returning Feedback**
  - The written comments will be returned to each group along with their grade.

MS From link to submit the grades:

One per group

https://forms.cloud.microsoft/r/3bhGjQ0cNf?origin=lprLink

Peer Grading Procedure

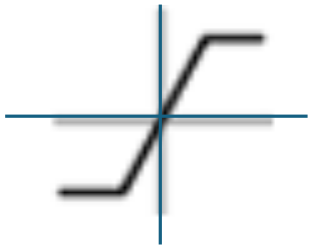
- Review the experimental design submitted by your peer.
- Evaluate how well the experimental setup, procedure, and data collection are explained.
- Assess whether the results could realistically be obtained and validated.
- Assign a score using the rubric below and provide one short written comment (strengths + improvement suggestion).

Criteria	Description	Points
Clarity of Experiment (2 pts)	Executable exactly as written, with no questions asked.	0–2
Parameter Identification (2 pts)	Determines both parameters independently.	0–2
Experimental Setup (2 pts)	Practical and safe; voltage and stall limits stated.	0–2
Data Collection & Method (2 pts)	Says how each quantity is measured, with repetitions.	0–2
Feasibility & Realism (2 pts)	Achievable with the available hardware.	0–2

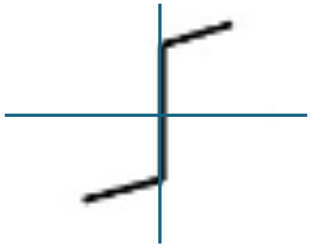
Total: /10

System Nonlinearities

saturation of the motor amplifier



friction in the motor

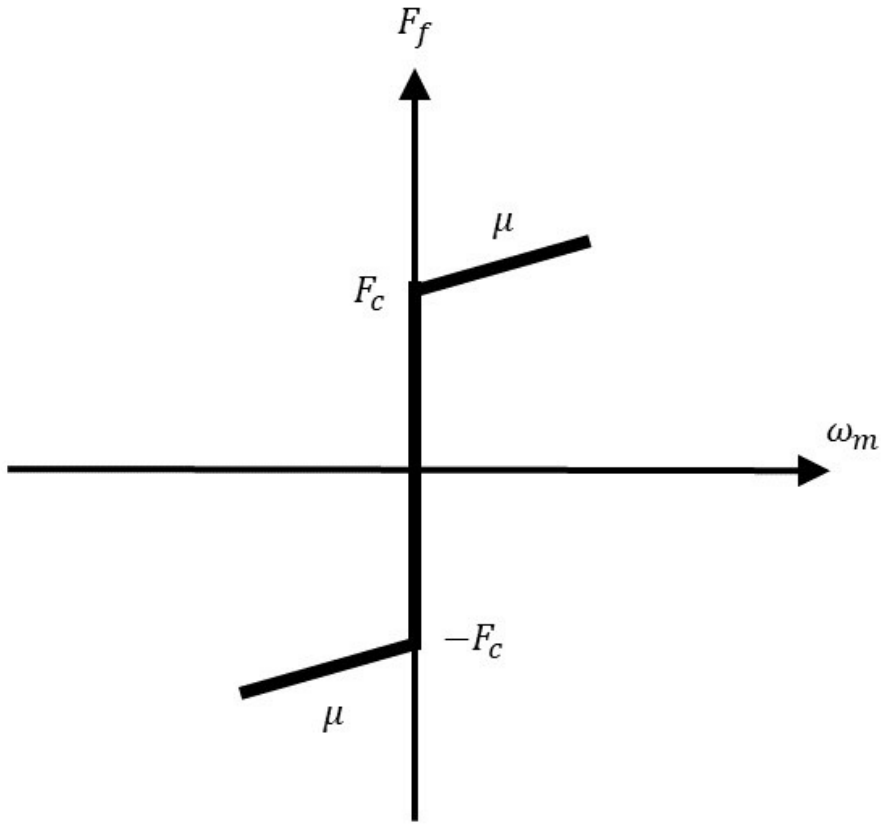


Friction Modeling

- In lab 2, the effects of friction weren't added to the system's models, but this reduces the accuracy
- The common friction model is the Coulomb (static) and viscous (dynamic) friction model
  - Linear with a discontinuity
- Friction Model

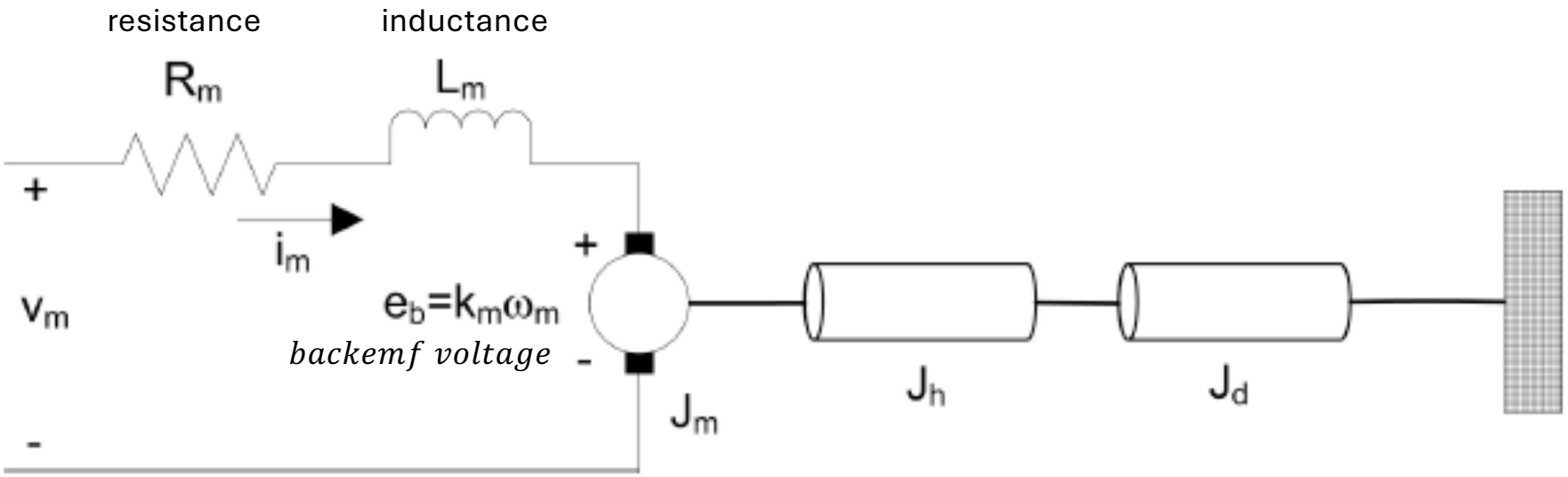
$$F_f(t) = \mu\omega_m(t) + F_c sgn(\omega_m(t))$$

$$sgn(x) = \begin{cases} -1 \; if \; x < 0 \\ 0 \; if \; x = 0 \\ +1 \; if \; x > 0 \end{cases}$$



F_c is the breakaway force (required to start moving)
ω_m is the angular velocity of the motor
μ is the viscous friction coefficient

DC motor


resistance
inductance
R_m
L_m
+
i_m
v_m
-
+
$e_b=k_m\omega_m$
backemf voltage
-
J_m
J_h
J_d

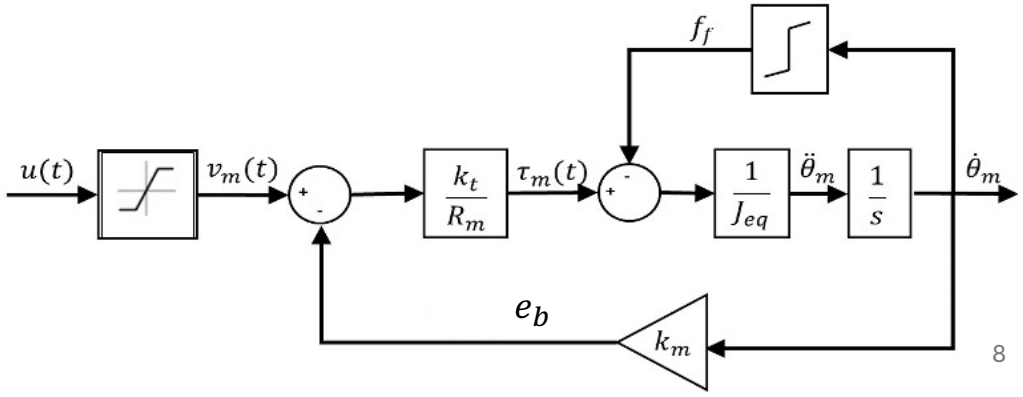
DC motor and load model with friction

Kirchoff's Voltage Law: $v_m(t) - R_m i_m(t) - L_m \frac{di_m(t)}{dt} - k_m \omega_m(t) = 0$

Assuming $L_m \ll R_m$: $v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$ Eq 1

Shaft with friction: $J_{eq} \frac{d\omega(t)}{dt} = \tau_m(t) - f_f(t)$ Eq 2

Applied current: $\tau_m(t) = k_t i_m(t)$ Eq 3



$backemf\ voltage$: $e_b = k_m \omega_m(t)$

Combining Eq 1,2,3:

$$J_{eq}\dot{\omega}_m(t) = \frac{k_t}{R_m}\left(v_m(t) - \boxed{k_m\omega_m(t)}\right) - \boxed{\mu\omega_m(t)} - F_c sgn(\omega_m(t))$$

backemf voltage

viscous friction

backemf voltage behaves as viscous friction!

Rearranging:

$$J_{eq}\dot{\omega}_m(t) = \frac{k_t}{R_m}\left(v_m(t)\right) - \frac{k_t}{R_m}k_m\omega_m(t) - \mu\omega_m(t) - F_c sgn(\omega_m(t))$$
$$= \frac{k_t}{R_m}\left(v_m(t)\right) - F_f',$$

Define F_f'

$$F_f' = (k_m + \frac{R_m}{k_t}\mu)\omega_m(t) + \frac{R_m}{k_t}F_c sgn(\omega_m(t))$$

- Since this equation is similar to $F_f = \mu\omega_m + F_c sgn(\omega_m)$, where μ is called viscous damping. Thus, $(k_m + \frac{R_m}{k_t}\mu)$ is called the equivalent viscous damping.

What you actually measure

Symbol	Meaning	Units	Where it comes from
k_m	back-emf constant	V·s/rad	datasheet (Lab 2, Table 1)
μ	viscous friction, torque form	N·m·s/rad	not measured directly
$\mu_f = (R_m/k_t)\mu$	viscous friction, voltage form	V·s/rad	from the slope of your fit
F_c	Coulomb friction, torque form	N·m	not measured directly
$V_c = (R_m/k_t)F_c$	Coulomb friction, voltage form	V	the breakaway voltage you measure

At steady state ($d\omega_m/dt = 0$) the motor equation reduces to

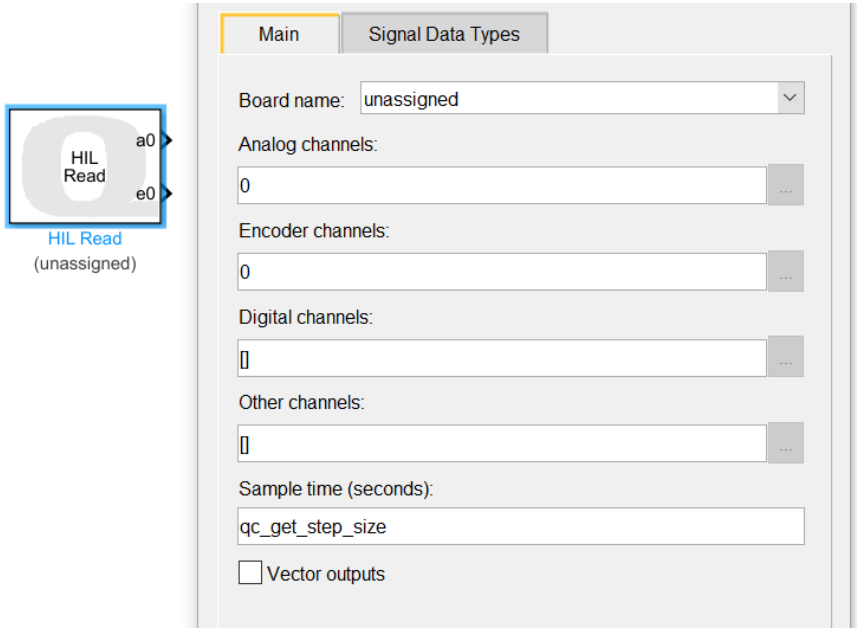
$$\mathbf{v_m = (k_m + \mu_f)\,\omega_m + V_c\,sgn(\omega_m)}$$

So the line you fit to V_m against ω_m has slope $(k_m + \mu_f)$ and intercept V_c. Therefore μ_f = slope − k_m.

If you get a negative value: you have most likely subtracted the datasheet k_m, an average over many motors, from a slope measured on your own motor. Report what you measured and say so. The k_m you identify in Section 6.3 is specific to your device — use that one instead. Never subtract μ (N·m·s/rad) from a slope that is in V·s/rad.

Qube Current Measurements

- Analog input 0 from the Qube reads the current running through the system
- Can be used at the same time as an encoder as shown to the right



Qube Velocity Measurements

- Convert counts to velocity (rad/s)
- Apply a first-order low-pass filter

Lab Procedure

- **Section 6.1 – Friction Identification**
- Identify the friction model parameters.

- **Section 6.2 – Model Friction Validation**
- Validate the motor model, including friction, using the identified parameters.
- Compare results with:
  - The model **without friction**
  - The **hardware response**

- **Section 6.3 – Motor Resistance and backemf Constant Identification**
- Identify **Rm**
- Identify **km**
- Derive the transfer function using these identified values
- Compare:
  - Model with identified parameters
  - Lab 2 model
  - Hardware response

Reference

- Quanser lab manual
- Feedback Control of Dynamic Systems (Franklin, Powell and Emami-Naeini)